

Sub phase -3 problem-solution

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Team ID	SWTID-2026-7624
Project name	Script-controlled ACL
Maximum mark	4 marks

1. Problem – Solution

1.1 Problem

In organizational and institutional environments, sensitive records may require different levels of access for different users. Standard Role-Based Access Control (RBAC) may not be sufficient when access needs to be restricted according to specific field values within individual records.

For example, records in the Institution Details table may belong to different branches such as ECE, EEE, and CSE. A user assigned to the EEE branch may need permission to view only EEE-related records.

The major problems identified are:

1. Standard role-based access may provide broader access than required.
2. Users may need access only to records matching a particular field value.
3. Creating multiple static ACL rules for different conditions can become difficult to maintain.
4. Different users require different CRUD permissions.
5. Administrators require unrestricted access to all records.

1.2 Solution

The proposed solution is to implement a Script-Controlled ACL in ServiceNow.

The system combines user roles with field-based conditions to make a dynamic access decision. The ACL checks the user's role and evaluates the required record condition before allowing or denying access.

The solution provides:

- Field-based record restriction.
- Role-based access control.
- Separate permissions for Read, Create, Write, and Delete operations.

- Full access for administrators.
- Dynamic access decisions using scripting.
- Better flexibility and maintainability.

1. Problem – Solution		Problem vs Solution (Summary)	
 Problem - Standard Role-Based Access Control (RBAC) is not sufficient to control access to every record. - Records may belong to different departments (e.g., IT, HR, Finance). - A user may need access only to records where a specific field (e.g., Department) has a particular value (e.g., IT). - Creating multiple static ACL rules for each condition is complex and hard to maintain.	 Solution - Implement a Script-Controlled ACL in ServiceNow. - Combine user roles with field-based conditions for dynamic access control. - Provide flexible and granular access to records based on the value of a specific field. - Reduce the number of static ACL rules and improve maintainability and security.	Aspect	
		Problem	Solution
		Access Control	Only role-based (RBAC)
			Script-controlled (RBAC + Field value)
		Record Filtering	Not possible
			Based on field value (e.g., Branch = EEE)
		ACL Management	Multiple static rules
			Single script-based ACL
		Security	Low flexibility
			Dynamic and granular access
		Maintenance	Difficult
			Easy and maintainable